

Group 76 Final Report: Classification of Retinal Diseases Using Optical Coherence Tomography (OCT) Images

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1 Introduction

Healthcare providers deal with many vision related illnesses such as Diabetic Macular Edema (DME), Choroidal Neovascularization (CNV), and drusen. The utilization of these retinal scans detect and determine these conditions. An issue that arises is the uniqueness of each ailment, as each condition varies in subtle ways. Correctly identifying retinal diseases can prove to be a challenge in a plethora of cases and with approximately 30 million OCT scans performed each year, this process can be time-consuming and labor-intensive for healthcare professionals. By having a classification model that can help speed up this process, healthcare professionals should be able to focus their attention on helping more individuals requiring their aid.

2 Related Work

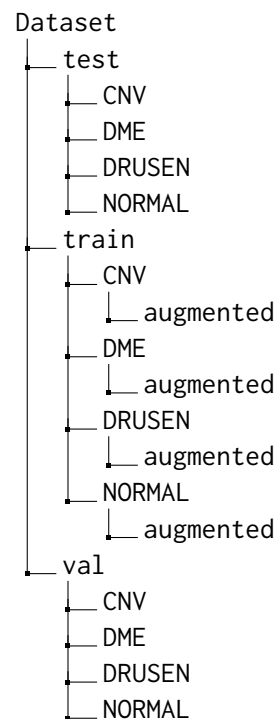
With the rising popularity of Machine Learning (ML) and Deep Learning (DL) in recent years, it is not surprising that many researchers around the world have worked on the classification of retinal diseases using ML and/or DL methods. Most related works have approached the problem by incorporating transfer learning to perform multi-class or binary classification (??).

As seen in the notebook created by the Kaggle user Badduri Nagendra Reddy (?), the fine-tuned model based on ResNet50 was able to achieve an impressive 98.97% final test accuracy on the same dataset as the one used in this project. Similarly, the models defined in the papers typically build upon a pre-trained model, where the authors only unfreeze the weights of the last few layers and add a final classification layer (??).

3 Dataset

3.1 Folder Structure

When downloading the data, the dataset was partitioned as follows:



The new added folders for the training data, augmented, contains the augmented image data for each category. The reason for this will be elaborated in the **Data Augmentation** section.

3.2 Initial Image Counts

Type	Disease	# of Images
train	CNV	37205
	DME	11348
	DRUSEN	8616
	NORMAL	26315
test	CNV	242
	DME	242
	DRUSEN	242
	NORMAL	242
val	CNV	8
	DME	8
	DRUSEN	8
	NORMAL	8

3.3 Data Augmentation

One of the encountered problems with the dataset is the class imbalance. As seen in the **Initial Image Counts** table, the number of images vary drastically. For example, CNV has nearly 4.5 times the amount of images the DRUSEN class has. Because of this, the model will be biased towards the classes with more images, thus resulting in poor performance. To mitigate this, data augmentation was performed on the training dataset to increase the number of images such that each class has the same number of images. This was done by performing the following random actions:

```
augmentation_transforms = transforms.Compose([
    transforms.RandomHorizontalFlip(),
    transforms.RandomRotation(50),
    transforms.RandomResizedCrop(size=224,
        scale=(0.8, 1.0)),
    transforms.ColorJitter(brightness=0.2,
        contrast=0.2),
])
```

This augmentation process includes randomly flipping the image horizontally, randomly rotating the image by up to 50 degrees, randomly cropping and resizing the image, and randomly changing the contrast and brightness of the image. By performing this, not only will the dataset be more robust as a whole, but the model will be less likely to overfit or have bias towards a certain class.

As a result, the new image counts after augmentation are as follows:

Type	Disease	# of Images
train	CNV	37205
	DME	37205
	DRUSEN	37205
	NORMAL	37205
test	CNV	242
	DME	242
	DRUSEN	242
	NORMAL	242
val	CNV	8
	DME	8
	DRUSEN	8
	NORMAL	8

This results in the training dataset having a total of 148820 images.

3.4 Preprocessing

Each image was loaded from the various folders as a PIL (Pillow, not the Python Imaging Library) Image and then converted to a tensor object. The pipeline varies slightly for the *train* dataset when compared to both *test* and *val* datasets. Both pipelines resized the images to 224 pixels by 224 pixels and normalized them with means [0.485, 0.456, 0.406] and standard deviation (std) of [0.229, 0.224, 0.225]. Since we are using ResNet50 as the base model, the values for the mean and standard deviation come from ImageNet's preset values, which is the dataset that ResNet is being trained on. The only difference in the preprocessing pipeline is that before normalization, the *train* dataset images might be horizontally flipped whereas the image data for the *test* and *val* pipelines is not. This allowed for the training set to have a more robust dataset. However, this is technically not considered an augmentation since no new data was created (no data distribution changes).

3.5 Annotation

Each image was labeled with an integer to represent its corresponding disease. The images were already named as DISEASE-#####-.jpeg and located in their corresponding folders, which allowed for ease of labeling. When assigning labels to the dataset, we use the following mapping:

CNV	0
DME	1
DRUSEN	2
NORMAL	3

4 Features

4.1 Data Input

Since this is a computer vision/image classification task, the features were derived from the pixels of the images. Each pixel was denoted by three values: Red(R), Green(G), and Blue(B). As mentioned previously, each image was cropped and normalized with values corresponding to the means and standard deviations for images from ImageNet.

4.2 Features

The selection of features was not done directly. This was because it was not required as there are no features to select. Since the pixels are the "feature", this would not be justified. As a result, feature engineering and feature selection was not needed. The same can be said for learning embeddings, as this is not a natural language processing task.

For the output features, there are four main features, which are the four categories of diseases: CNV, DME, DRUSEN, and NORMAL.

5 Implementation

5.1 Learning Model

Instead of training a model from scratch, a pretrained neural network was used and adapted it to the use case. The pretrained neural network selected was ResNet-50, which was trained on the ImageNet dataset. ResNet-50 is a neural network that is 50 layers deep originally developed by Microsoft Research in 2015. This is a type of convolutional neural network (CNN), which is a subset of neural networks that are used for image recognition.

5.2 Implementation Steps

5.2.1 Initializing the Neural Network

To implement the ResNet-50 based neural network, the first step was to instantiate a neural network that uses ResNet-50. This occurs after all the images have been preprocessed. This was called CustomResNet. For the initialization, the input features and classifiers are all defined. The initialization was very similar to typical setup for neural networks.

5.2.2 Freezing and Unfreezing Layers

Upon initialization, the first thing was to freeze all the layers but the last four layers, which includes three feature sequential layers and the classifier

layer. This was because a pretrained model such as ResNet-50 already comes with various useful general features such as edges, textures, and colours in early layers. Since the dataset that was used was quite small (four disease classes), training all the layers can result in an increase of time, overfitting, and wiping out pretrained knowledge.

5.2.3 Hyperparameter Tuning

While the developed model achieved respectable results during the progress report check-in, it left a great margin for improvement in the form of model optimizations and hyperparameter tuning, which were completed in this final stage. To begin, due to the size of the ResNet-50 and the large quantity of data, using Grid Search to narrow down the hyperparameter values, like the learning rate to fine-tune, would have been extremely costly in terms of runtime and computation power, making it not ideal. Therefore, learning rate scheduling was used to help fine-tune the model, as having a large starting learning rate value can dynamically decrease the learning rate in the context of the model's performance during model training. This allows for the model to arrive at the optimal learning independently while training on the large dataset without requiring the various runs that a fine-tuning operation like grid search would take.

The learning rate scheduler CosineAnnealingLR was chosen because it decreases the learning rate according to a cosine curve, which allows for an overall smooth learning rate curve that keeps training stable, allowing for better generalization in the model. As well as the ability to rapidly decrease the learning rate in the beginning when it is usually still far from the optimal value, and slows down near the end when it nears the optimal value, allowing for faster convergence. In addition to the learning rate scheduler, a prebuilt warmup scheduler function, GradualWarmupScheduler, created by a Kaggle user, was implemented. The warmup scheduler uses the first 5 epochs of the model as a kind of 'warm up', increasing the training rate slowly to the desired value, after which it hands it off to the normal Cosine scheduler to continue the learning rate scheduling. This warm-up helps stabilize the training of the model, which in turn allows for overall better generalization within the model, leading to better performance, and allows for a high beginning learning rate value to be set for the learning rate scheduler, which gives way for faster convergence. Lastly, to help further prevent overfitting,

save computation time, and achieve faster convergence, early-stopping was implemented with the monitoring of the validation loss with a threshold value of 0.0001 and patience of 10.

5.2.4 Criterion

The criterion function remained largely untouched from the progress report, as it was CrossEntropy-Loss, which was already well-suited to the image classification task seen here. It did receive a minor modification in the form of the `label_smoothing` parameter being set to 0.1. This regularization technique helped improve the overall generalization ability of the model, helping make the calculated target value of the loss become a mixture of the original ground truth value and a uniform distribution.

5.2.5 Optimizer

The optimizer was completely overhauled, with the Adam optimizer being replaced with Stochastic Gradient Descent (SGD). The reason for this change is that Adam operates better when the dataset quantity is lower, as it can converge quickly on smaller complex datasets and produce respectable results with less fine-tuning. Therefore, in the case of the progress report, this was ideal as it was only dealing with a fraction of a dataset; hence, an optimizer that is able to find the important generalized trends/patterns in a smaller amount of data was ideal. However, since the model is being fed the entire dataset in this final stage, Adam falls short, as its main disadvantage is generalization with a larger dataset, not allowing for the best output to be achieved. Therefore, a switch to SGD was clear, as it enables better generalization for image classification tasks on a large dataset, as the one seen here.

Compared to Adam, SGD required more tuning to truly operate at its best, and thus, it was configured with a starting learning rate value of 0.01 that was narrowed down through various thorough tests and is dynamically scaled according to the learning rate scheduler, `CosineAnnealingLR`. A momentum value of 0.9 that allows for faster convergence and generalization, as it increases the speed of the model when it is heading in the right direction, while slowing down when going in the wrong direction. It was also equipped with a weight decay value of 0.0001, which helped prevent overfitting by penalizing extremely large weights. Lastly, the optimizer model was configured to enable

Nesterov momentum, which boosts momentum and speeds smoother convergence by allowing the function to make better-informed updates as it steps using momentum, thus allowing it to not overshoot when arriving at the optimal solution.

5.2.6 Training Data

The dataset has 83484 images in the four folders (CNV, DME, DRUSEN, NORMAL) under training. The distribution was not even, as CNV had significantly more images compared to the other diseases. When comparing CNV to the category with the least images, DRUSEN, CNV has around 2.5 times the images in DRUSEN. This was remedied by augmenting images for each section to match CNV. This totaled 148820 images for model training, with 37205 images per category. By augmenting the images to even out the categories, this prevented skewing and bias.

5.2.7 Validation and Test Data

Since there are significantly fewer images categorized under these, all images were used. There are only 32 images for validation spread evenly in the four diseases. The testing data has an even spread too, with 242 images for each disease, totaling 968 images.

5.2.8 Training and Validation

For training, a total of 100 epochs were performed for the 148820 selected images. The loss and accuracy of both training and validation were printed along with a graph displaying the loss for training and validation.

5.2.9 Testing

After testing, the values were plotted on a 4×4 confusion matrix. The final accuracy of the model was written above it.

6 Results and Evaluation

6.1 Training/Validation Phase

After the model was initialized, it was fed through the three stages of training, validation, and testing, with the training and validation phases being combined in a singular loop to save computation time as well as allow for better performance metric representation. During the training and validation phase, the model's performance was measured via the computation of training/validation loss and training/validation accuracy at each epoch. This allows insight into the performance trend of the

model and its overall journey through the training and validation phase. The rate at which the model learns the trends in the dataset can be witnessed in its ability to better classify the images through the loss and accuracy metrics. The loss method was the cross-entropy loss, which is accessed through the `nn.CrossEntropyLoss()` function that preexists as a part of PyTorch, and the accuracy was calculated via the division of the correct predictions by the model on both the training and validation datasets by the total predictions by the model on the training and validation sets, respectively. In conjunction with this, loss values are stored and calculated for the training and validation steps, and then visualizing the overall trends by plotting them on a line graph to help further understand the model and its performance.

6.2 Test Phase

By this stage, the model has been evaluated once through the validation phase; however, it was evaluated again, this time on the test split of the dataset via the testing phase. For the testing phase, the performance metric of accuracy was used as the evaluation metric of the performance of the model, which was calculated via the same accuracy calculation method as training and validation. In addition to these performance metrics, a confusion matrix was displayed, which visualizes the true positive (TP), true negative (TN), false positive (FP), and false negative (FN) predictions of the model for the testing phase. This aided in easier comprehension and evaluation of the performance of the model on the testing dataset, thus presenting better methods of improvement.

6.3 Quantitative Results

The following results are shown below:

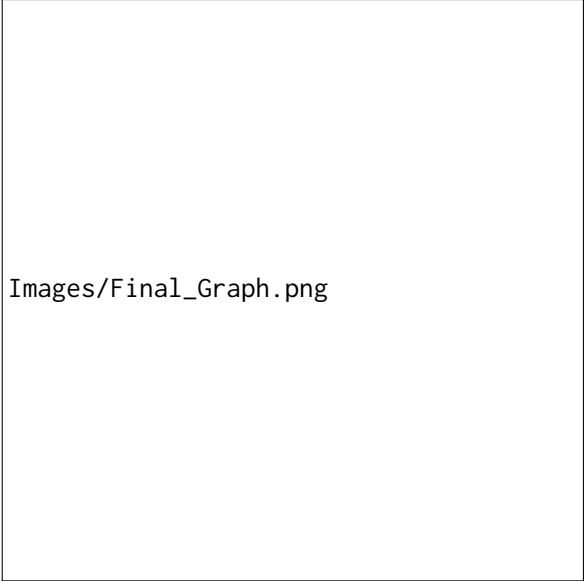


Figure 1: Graph of training and validation loss over epochs from the final model.

Metric	Value
Accuracy	99.6901%
Precision (Macro)	0.9969
Precision (Micro)	0.9969
Recall (Macro)	0.9969
Recall (Micro)	0.9969
F1 Score (Macro)	0.9969
F1 Score (Micro)	0.9969



Figure 2: Confusion Matrix of the test results from the final model.

6.4 Baseline and Previous Results

6.4.1 Baseline Results

The baseline mode that was used for a comparison was a random classification model. This model randomly selects a class for each image without

any learning or training. As a result, the expected accuracy for this model is 25% since there are four classes.

Metric	Value
Accuracy	25.31%
Precision (Macro)	0.2527
Precision (Micro)	0.2531
Recall (Macro)	0.2531
Recall (Micro)	0.2531
F1 Score (Macro)	0.2526
F1 Score (Micro)	0.2531

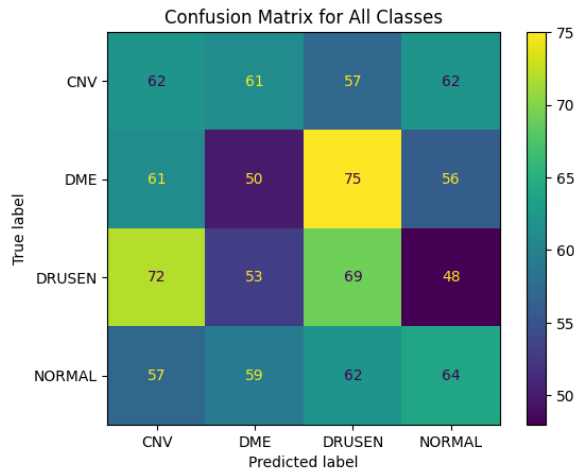


Figure 3: Confusion Matrix for all classes using the random classifier.

Adding onto this, each of the classes were individually evaluated. The results are shown below:

Class	Precision	Recall
CNV	0.25	0.26
DME	0.22	0.21
DRUSEN	0.26	0.29
NORMAL	0.28	0.26
Macro Avg.	0.25	0.25
Weighted Avg.	0.25	0.25

Class	F1-score	Support
CNV	0.25	242
DME	0.22	242
DRUSEN	0.27	242
NORMAL	0.27	242
Accuracy	0.25	968
Macro Avg.	0.25	968
Weighted Avg.	0.25	968

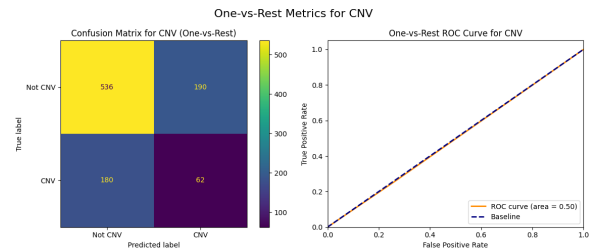


Figure 4: Confusion Matrix for CNV class using the random classifier.

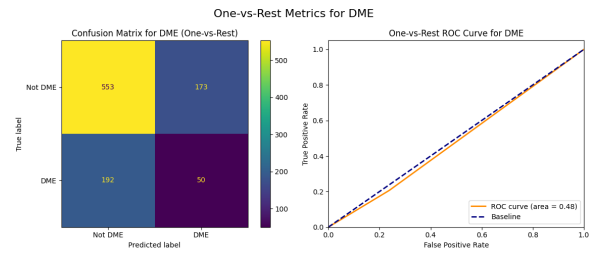


Figure 5: Confusion Matrix for DME class using the random classifier.

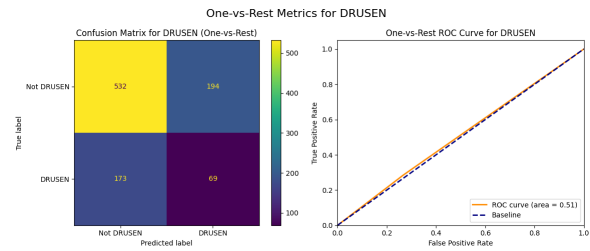


Figure 6: Confusion Matrix for DRUSEN class using the random classifier.

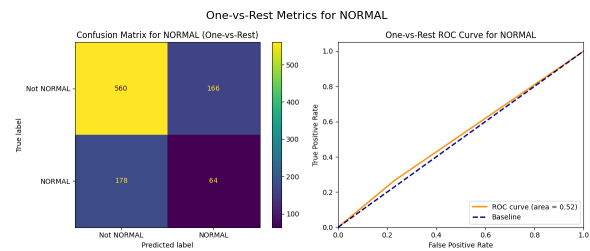


Figure 7: Confusion Matrix for NORMAL class using the random classifier.

6.4.2 Previous Results

These are the results from the progress report. The results are not as comprehensive since many metrics such as precision and recall were not added.

Metric	Value
Accuracy	84.40%

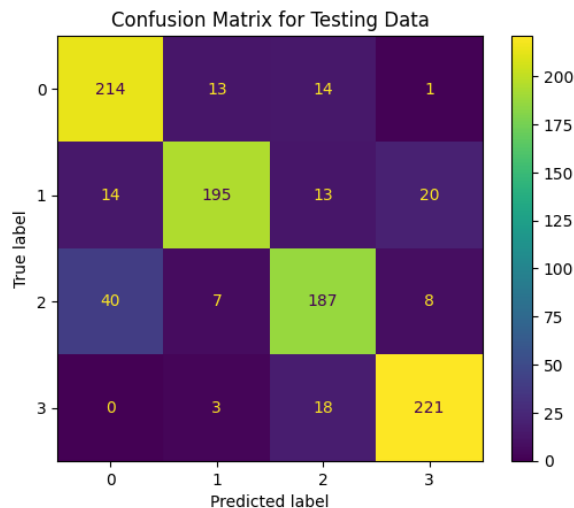


Figure 8: Confusion Matrix from the progress report.

7 Progress

7.1 Feedback and Planned Improvements

From the progress report, there were a few problems with the model and the implementation that were highlighted. To start off, the dataset for the model was imbalanced. While this was not important for the progress report since 1000 images were used for each category, this would be a problem when scaling up to a full dataset. The second issue was the lack of a baseline comparison such as Majority Class or a simple CNN. This leads into the third problem, the lack of additional metrics such as micro and macro F1 score, recall, and precision. Lastly, there was a lack of hyperparameter tuning and model optimization. There was a lack of momentum in the optimizer, and the learning rate was not adjusted throughout training.

7.2 Implemented Improvements

To address the issues highlighted in the feedback, the following improvements were implemented:

7.2.1 Data Augmentation

As highlighted in section 3.3, data augmentation was performed to ensure a balanced dataset. Images in the other three categories (DME, DRUSEN, NORMAL) were augmented to match the number of images in CNV, which included random horizontal flips, random rotations, random resized crops, and color jittering.

7.2.2 Additional Metrics

The additional metrics were calculated to better evaluate the model's performance. These metrics include precision, recall, F1 score (both micro and

macro), which were printed with correspondence with the confusion matrix. These metrics provide a more comprehensive understanding of the model's strengths and weaknesses across different classes.

7.2.3 Baseline Comparison

To address the lack of a baseline comparison, a random classifier was used as a baseline. This classifier

7.2.4 Hyperparameter Tuning

To help the model achieve better performance, several model optimizations and hyperparameter tuning were conducted. Including ones that involved planning, like the addition of momentum in the optimizer and learning rate adjustments and tuning in the form of the learning rate scheduler, and ones that were not planned, like the transition from the Adam optimizer to SGD, the early-stopping, and the unfreezing of the last three layers of the ResNet-50 model, in addition to the classifier. The reason for the extra additions is that research during the fine-tuning and optimization process revealed improvements that not only improved the model's generalization, leading to better performance, but also accelerated convergence, which was a priority given the dataset's massive size, resulting in long computation time. Furthermore, as the Applications of Machine Learning course progressed, more information regarding ways to improve a model was discovered that helped influence the optimizations and additions made to the model. For a full breakdown of the hyperparameter tuning and model optimizations implemented, as well as the reasoning behind the choices, please refer to section 5.2.

7.3 Progress Evaluation

When viewing all the feedback and the implemented improvements, all the proposed feedback and plans were implemented. There was no change in course of action because the final test accuracy for the model was 99.69%, thus hitting the initial target proposed.

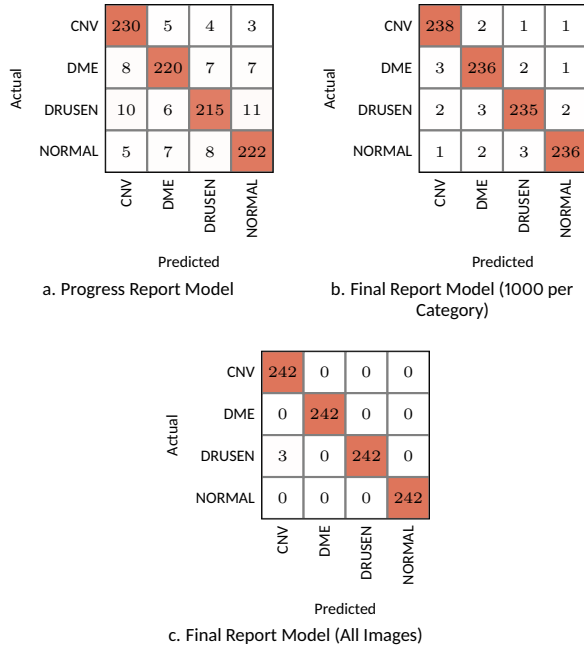


Figure 9: Confusion matrices for three models generated by group 76: Progress Report Model, Final Report Model (1000 per Category), and the Final Report Model (All Images).

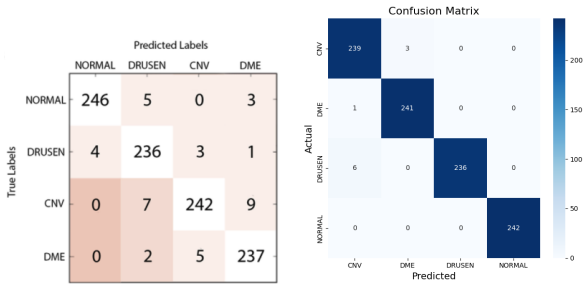


Figure 10: Confusion matrices from models outlined in (?) and (?).

8 Error Analysis

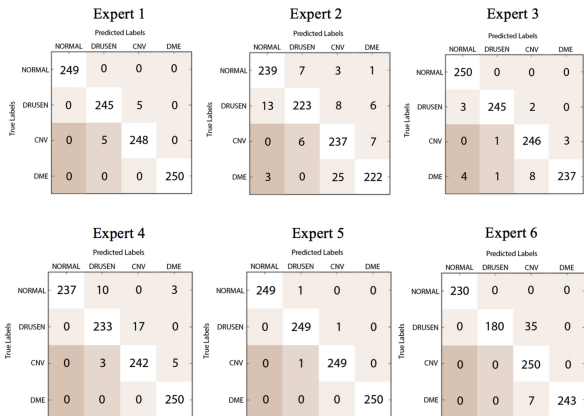


Figure 11: Confusion Matrix from the progress report.

As shown in section 6.3 and figure 9, the final model was able to perform incredibly well, only

making 3 wrong predictions out of all 968 data points in the test set. It was also able to reach a perfect 1.0 recall, precision, and f-1 scores for the NORMAL-vs-Rest classification. This means that all cases of a retinal disease would be classified as a disease, which will help the early detection of retinal diseases and hopefully prevent any condition from being undiagnosed. Comparing this to the results shown in Kermany et al. (?) and Reddy (?), it can be seen that the model seems to outperform most if not all of these implementations, including human experts.

On the other hand, figure 9 clearly shows that the final results for the model trained on all the images has only 3 incorrect predictions, all of which come from DRUSEN images that were classified as CNV. This is a pattern that can be seen throughout the different versions of the model's implementation and in the results from Kermany et al. (?) and Reddy (?) as seen in figures 10. This bias towards predicting CNV could be caused by the distribution of the training dataset itself which has a lot more CNV images. While data augmentation was performed to get DME and DRUSEN's data to be the same amount as CNV, these newly added data were essentially slight modifications to the original and much smaller dataset. Thus, it is possible that the model had a harder time learning the features of DME and DRUSEN when compared to CNV. However, it can be seen in figure 11 that human experts also tend to mostly misclassify a DME and DRUSEN as CNV, which could suggest that the bias that is seen in the model is a result of the characteristics of the underlying problem itself.

One thing that can be done to possibly improve the model and mitigate this bias is to downsample the majority class, which in this case is the CNV class. By downsampling it, there will be less data augmentation required for the other classes such as DME and DRUSEN. An experiment can be conducted to see if different distributions of augmented vs non-augmented data change the bias of the predictions that the model makes. However, it must be noted that while downsampling the dataset might fix the bias issue, it is also important to keep a large enough dataset for each class so that the model can still learn the required features to perform its task.

9 Team Contributions

9.1 Elite

- Worked on the initial image opening and storing of data.
- Implemented the image augmentation process.
- Wrote sections 1, 3, 4, and 5 of the document and worked on overall presentation of Latex

9.2 Ishpreet

- Wrote the model training code and the ResNet-50 neural network.
- Trained and tested the model.
- Wrote section 6 of the document.

9.3 Kenneth

- Worked on setting up the neural network and research for the start of the project.
- Wrote sections 2 and 7 of the document.

References

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